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Development of an ISOBUS-compliant communication node for multiple machine vision systems on wide boom sprayers for nozzle control in spot application schemes

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ABSTRACT

This study focuses on developing a node named machine vision node (MVN) for hybrid communication between Ethernet and Controller Area Network (CAN) on the implement bus of boom sprayers. This enabled the integration of multiple machine vision systems and simultaneous control of as many as 60 nozzles based on pest detection results by machine vision. The MVN consists of an electronic control unit (ECU) built around a single-board computer, equipped with a 2-channel CAN hat and an Ethernet router. The ECU utilizes AgIsoStack++ library to maintain ISOBUS compliance, enabling data exchange between the MVN and other electronic components on the sprayer. A threaded client-server firmware incorporates a checksum verification subroutine to ensure reliable protocol message handling, followed by message queuing for uninterrupted real-time processing. The MVN demonstrated robust communication and processing capabilities while receiving protocol messages via Ethernet, reading speed-related CAN frames, and converting CAN data for individual nozzle control simultaneously. The MVN parses and synchronizes over 30 predefined protocol messages every 40 ms. The CAN bus load on the implement bus increased by only 5.86 %, which remained well within the accepTable 45 % limit. ISOBUS compatibility tests across three different virtual terminals confirmed interoperability and standardized control within 51.5 to 70 s. In-field tests confirmed that the MVN dynamically adjusted nozzle opening times based on vehicle speed, maintaining consistent spray lengths of 71.52 cm across different speeds of 3.22 kph, 6.44 kph, and 9.66 kph with nozzle opening times ranging from 270 ms to 800 ms.

1. Introduction

Modern machinery across different industries increasingly relies on advanced electronic systems to automate equipment while ensuring high levels of reliability, improved performance, enhanced safety, and increased precision in operations [1]. In the automotive sector, for instance, electronic control systems have evolved from basic operational support to managing complex processes such as adaptive cruise control, automated braking, and autonomous navigation [2]. In agriculture, the automation of machinery such as tractors, sprayers, and harvesters has utilized electronics that significantly improve productivity while also reducing environmental impact [3].

Automation often involves the integration of electronic control units (ECU), sensors, and actuators to enable precise control and monitoring [4]. Each ECU consists of chipsets, hardware components, and embedded software to serve as an interface between sensors and actuators. In a tractor, agricultural implements such as plows, seeders, planters, cultivators, sprayers, and harvesters may incorporate their own set of ECUs to communicate within the machine or across other machines [5]. Nowadays, implements may range from simple trailers to complex machinery equipped with advanced technologies like artificial intelligence-based automatic detection for autonomous driving to actuation control [6].

As electronic systems in agricultural machinery become increasingly complex, the adoption of standardized communication protocols has facilitated efficient data exchange among components contributing to

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field efficiencies ranging from 68 % to 71 % in operations such as planting, field cultivation, and anhydrous application [7]. Therefore, in modern agricultural vehicles, a standard message-based serial communication protocol known as controller area network (CAN) has become the industry standard due to its reliability, reduced design complexity, cost-effective scalability, and the ability to facilitate real-time communication between sensors and actuators [8,9]. When multiple ECUs, sensors, and actuators send and receive messages through CAN, the system is referred to as a CAN bus which was standardized across different manufacturers and industries [10]. CAN follows the open system interconnection (OSI) layer model, in which its data link layer, known as CAN 2.0B, is standardized by ISO 11,898 [11], ensuring that devices can communicate regardless of the system or application. This standardization specifies message identification and prioritization so that each message is properly tagged with its source and intended recipient [12]. At the application layer, several protocols have been developed to serve different industries including ARINC825 for airborne systems [13], CAN fieldbus for automotive [14], CANopen for embedded control [15], DeviceNet for industrial automation [16], NMEA2000 for marine systems [17], SAE J1939 for heavy-duty vehicles [18], and ISO11783 (colloquially called ISOBUS) for agriculture and forestry [19]. ISOBUS is a key building block for advanced agricultural practices [20], enabling seamless communication between equipment from different manufacturers and managing complex precision farming tasks [21].

Nevertheless, despite the distinguished protocols at the application layer, CAN allows communication between these protocols if they share the same data link layer. For instance, at the implement connecting a tractor with an implement various application link protocols can be found including ISOBUS itself to control the implement, J1939 to handle engine data, and NMEA 2000 to encode global navigation satellite system (GNSS) information [22]. In fact, the data distribution of an implement bus would typically include approximately 50 % from ISOBUS, 24 % from NMEA 2000, 16 % from J1939, and 10 % from proprietary manufacturer-specific data [23]. One important feature in ISOBUS is the virtual terminal (VT), which is a universal interface between the machine and operator, allowing control of different implements from a single display. Also, the task controller (TC) automates data collection and reporting, integrating farm management systems [24,25]. Thirdly, the tractor implement management (TIM) further enhances safety by enabling bidirectional communication between the tractor and the implement for smoother, more automated operations [26].

Paraforos et al. [27] reviewed 73 research articles on ISOBUS-based agricultural implements and sensors and observed a focus on the implementation and application of ECUs and sensor systems but a significant shortage of insights into the ECU integration processes. Nevertheless, few researchers have acknowledged this topic including Doopalam et al., [28] for wireless analyzers of ISOBUS, Suomi & Oksanen [29] for seed drill depth control, Sharipov et al. [30] for centrifugal disc spreaders, [31] for auxiliary sensors, and Al-Mallahi et al. [32] for sprayers.

In their study, Motalab & Al-Mallahi, [33] developed an ECU to integrate machine vision systems with boom sprayers, enabling individual nozzle control for pesticide spot application. Although their ECU could handle machine vision with multiple cameras, it was limited to communicating with a single machine vision system, which limited its scalability towards wide boom sprayers that require multiple machine vision systems with dozens of cameras. While – theoretically – adding multiple ECUs (one for each machine vision system) is possible, the implement bus has a capacity limit of 45 % ([34]; Kvaser, 2023) to prevent communication errors. Therefore, more complex ECUs integrate several sensing systems before reaching the implement bus may be needed, serving as sensor nodes [35,36]. Similarly, wide boom sprayers, which require multiple machine vision systems necessitate ECU designs that consolidate data from various sources into a single node for efficient communication and compatibility with the implement bus.

As such, this study presents the development of a novel machine vision node (MVN) that integrates multiple machine vision systems via a hybrid communication network consisting of CAN and Ethernet. Unlike conventional systems requiring independent ECU for each machine vision system, this design allows for timely and simultaneous control of as many as 60 nozzles using any number of cameras that cover the width of a sprayer boom. To ensure robust data integrity, the MVN incorporates a checksum verification subroutine, enabling reliable protocol message handling coming through Ethernet. Also, the threaded architecture of the client-server program provides continuous real-time message processing. At the CAN level, the MVN was developed to achieve full ISOBUS compliance through the open-source AgIsoStack++ library, enabling the MVN to self-register on the ISOBUS network. Hence, the MVN interacts with the CAN bus of the sprayer to receive speed data via CAN frames and adjusts the nozzle openings according to the speed of the sprayer. By addressing key limitations of existing systems, this MVN represents a significant step forward in enabling scalable, precise, and resource-efficient spraying solutions for wide boom sprayers.

2. Materials and methods

2.1. Hardware of MVN

The MVN depicted in Fig. 1 consisted of an electronic entity for synchronized and simultaneous management of two different communication protocols: Ethernet and CAN. The ECU was implemented using a single board computer (SBC) (Raspberry Pi 4B, Raspberry Pi, Milton, Cambridge, United Kingdom), powered by an ARM Cortex-A72 processor. The SBC was chosen for the ECU platform because of its Ethernet interface and the ability to add CAN interfaces. The SBC was crowned with a 2-Channel CAN bus expansion HAT (Waveshare, Shenzhen, China) that supported CAN2.0. Raspbian, as its operating system, supported the necessary multithreading to manage concurrent processes. Ethernet of the SBC was connected to a router (Asus AC1750, Taipei, Taiwan) and received protocol data from multiple vision systems to make a fully functional node MVN. The two-channel isolated CAN bus interface ensured communication with ISOBUS networks. The first channel, i.e., CAN 0, was connected to the implement bus of the sprayer Case IH Patriot 3340 (CASE International, Wisconsin, United States) to read speed data coming from the tractor bus through the tractor ECU (TECU) to the sprayer implement bus. The second channel of the CAN HAT, CAN 1, communicated with another implement bus for the Pentair rate controller (Hypro ProStop-E ISOBUS System, Pentair plc, London, UK), which controlled 60 nozzles deployed parallelly with the original spraying boom and valve by the Raven rate controller (Raven Industries Inc., South Dakota, United States) of the CASE IH. The ECU was powered by a DC-to-DC voltage converter (CPT, Currents Logic Group Limited, Shenzhen, China), which received twelve volts from the tractor and supplied five volts to the SBC.

2.2. Software of MVN

In the system flow diagram detailed in Fig. 2, the system works in four stages: Initialization, Queue, Thread 1: to create nozzle control information, and Thread 2: to create and send CAN frame. At the initialization stage, the ECU sets itself up as a server by taking a fixed server port and internet protocol (IP) address within the router IP address class. All machine vision computers are assigned IP addresses as clients through the dynamic host configuration protocol (DHCP) by the router. The server continuously accepts new client connections at any time and always listens for incoming connections on the same IP address class. After any successful detection, the machine vision computers (i.e. clients) broadcast protocol messages on the Ethernet network and the ECU acting as a server receives all the messages via the router.

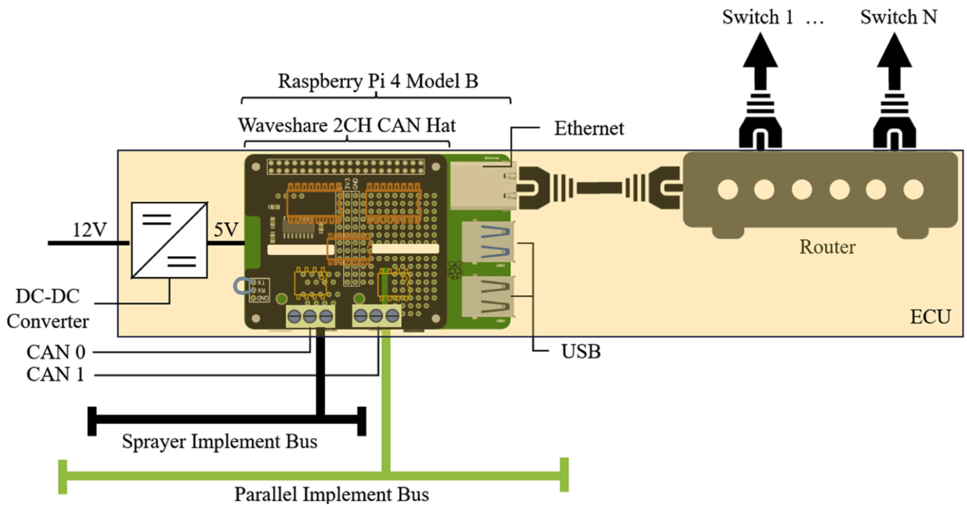


Fig. 1. MVN hardware consisting of SBC (Raspberry Pi) and two-channel CAN HAT, CAN 1 communicates with parallel implement bus, CAN 0 receives speed data from the sprayer.

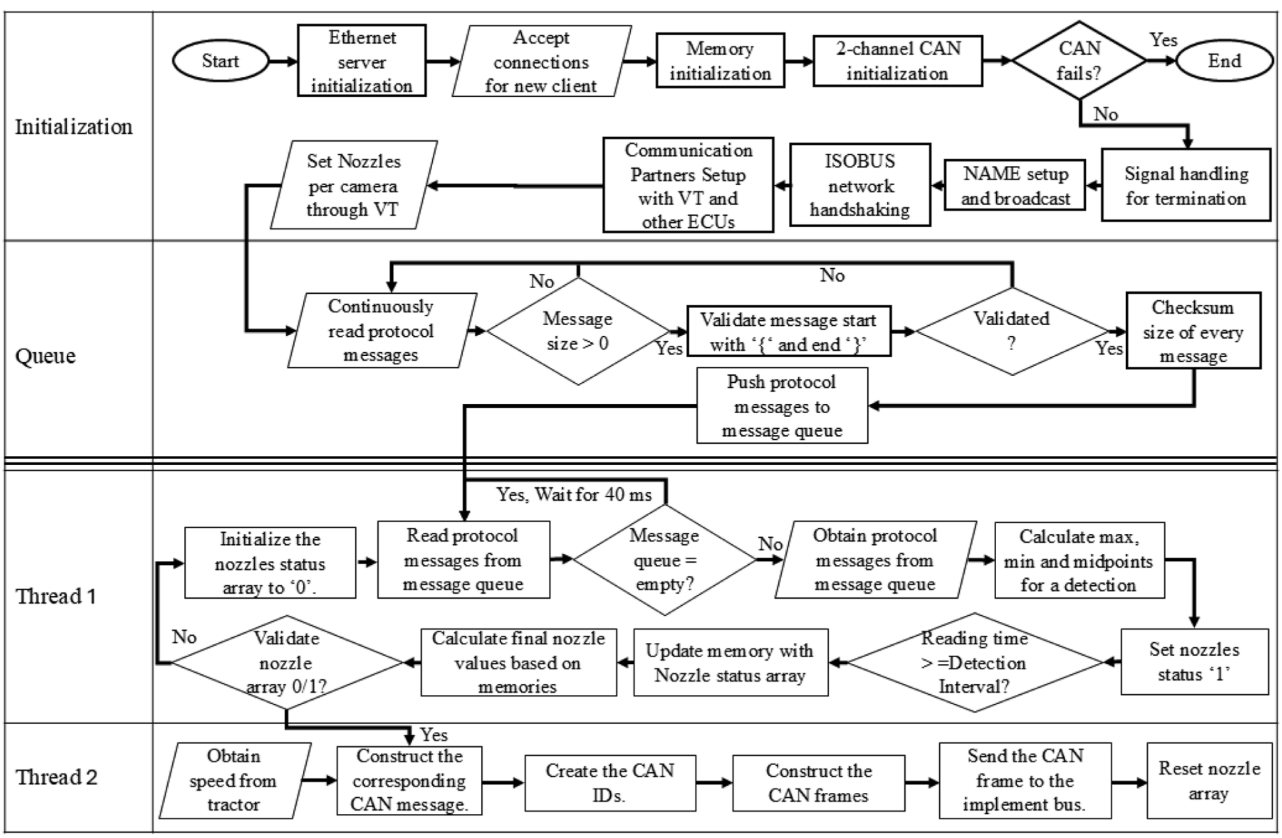


Fig. 2. Flow diagram of the MVN including the thread control for protocol messages and CAN frames.

Afterwards, the CAN network is configured to handle CAN bus communication, ensuring that all parameters are set for seamless network communication. The MVN memory storage also gets initialized to keep certain numbers of CAN frames as a temporary record. Utilizing the AgIsoStack++ library, the MVN registers itself on the ISOBUS network using a manufacturer ID of 0 × 1C, which corresponds to a registered but inactive manufacturer listed under the agricultural industry electronics foundation (AEF). In case 0 × 1C is unavailable, the system tries IDs 0 × 1D, 0 × 1E, or 0 × 1F as alternatives. NAME is a 64-bit identifier that uniquely distinguishes a control function within the network parameters [37]; in this case, it is specifically defined as an ISOBUS section control (SC) function which manages individual implement sections such as nozzle sections on a sprayer. These sections were controlled by individual cameras for precise spraying operations.

Following signal handling, the system broadcasts the MVN NAME, facilitating ISOBUS network handshaking and establishing communication partnerships with the VT and other ECUs. A user interface for the VT (Fig. 3) designed using the demo version of ISO-Designer v5.6.1 (Jetter AG, Ludwigsburg, Germany) allows the operator to directly adjust nozzles per camera. After successful handshaking, messages with

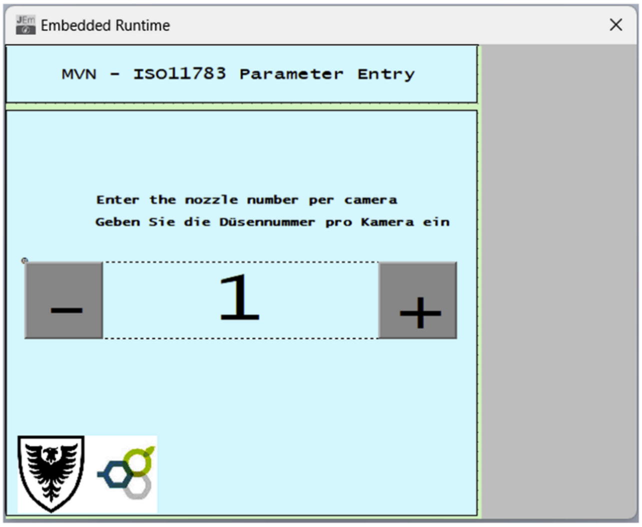


Fig. 3. Custom user interface for VT designed to enter the number nozzles controlled by one camera.

source or destination IDs made the VT screen visible, enabling the operator to control camera and nozzle ratios through the ISOBUS-compatible VT.

The Queue stage continuously waits for 42-character fixed protocol messages from any of the clients after the foundational setup is complete. These messages contain information from the cameras of the machine vision system, including Camera ID, Pest ID, pest location in the field of view (FOV) of the camera, total FOV width, time of detection, and detection completion time. An example protocol message looks as follows: {01,B,−0216,+0243,1000,1,623,850,684,951,0250} which was developed initially by Motalab & Al-Mallahi [33]. However, a checksum mechanism was additionally implemented in the algorithm to address the low fault tolerance of the Ethernet protocol by verifying the start and end of the message ("{" and "}") and calculating its length. The checksum gets recalculated and compared with the transmitted checksum to confirm that no errors have occurred during transmission. Upon receiving a complete message, all the protocol messages are immediately pushed into a queue.

Thread 1 stage continuously monitors the queue for new messages (Fig. 4). These messages are processed following the first in first out (FIFO) principle. Every 40 ms, each message in the queue would be processed to build a nozzle array ($\boldsymbol{A}$), which governs which nozzle to open across the boom sprayer.

Thread 2 reads and validates the nozzle array and includes them in the memory. In this research work, the MVN was programmed to control up to 64 nozzles of a Pentair rate controller. Since each byte in the CAN data frame controls two nozzles Pentair HYPRO [38], Eq. (1) was constructed so that,

$$b_k = (A[2k] \times 10) + (A[2k-1] \times 160), \; k = \{1, 2, \ldots, \; 32\} \tag{1}$$

where b, $\boldsymbol{A}$, and k represent one byte in the CAN data frame in decimal value, nozzle array, and nozzle index, respectively. Afterwards, Thread 2 creates 4 CAN IDs based on Eq. (2) starting from the hexadecimal value of 0 × 18BB961C,

$$G_d = 0 \times 18BB961C + (d \times 0 \times 100), \; d = \{1, 2, 3, 4\}, \tag{2}$$

where G and d represent the CAN ID and its order, respectively. These four CAN IDs with their date are formatted into CAN frames which get broadcast over the ISOBUS, ensuring seamless communication within the network.

Thread 2 includes an additional feature to adjust the nozzle actuation time based on the tractor speed to ensure that a nozzle would cover the same spraying area and density regardless of speed. To identify the tractor speed, it was required to get the appropriate parameter group number (PGN) from the CAN network, as the PGN represents a specific function of the CAN frame. In ISOBUS-compatible tractors, speed data could be found in several PGNs: the Ground-based Speed and Distance (PGN 0xFE49, e.g., CAN ID 0 × 18FE49FE), the Wheel-based Speed and Distance (PGN 0xFE48, e.g., CAN ID 0 × 18FE48FE), and GPS-based Vehicle Direction/Speed (PGN 0xFEE8, e.g., CAN ID 0 × 18FEE81C), etc. Among these, PGN 0xFE49 was selected to obtain the speed data, as it was the most commonly available on agricultural tractors. The speed data was found in the first and second quadrants of the CAN data and was converted by multiplying by 0.001 per bit to obtain the speed in m/s. When this PGN was received on channel CAN 0 of the CAN expansion HAT, the ECU converted the message to speed data. As a nozzle opening time of 400 ms was found to be appropriate for optimal spraying at a speed of 6.44 kph [33], Eq. (3) was derived to calculate the nozzle opening time,

$$t = \mathrm{C}\left(\frac{1}{v}\right), \tag{3}$$

where t and v are the nozzle opening time in ms and the speed of the sprayer in km/h, whereas C is a coefficient equals to 2576 ms.km/h. Finally, the program incorporated signal handling for smooth termination and confirmed that all processes were cleanly shut down upon program exit.

2.3. Lab apparatus and scenarios

To evaluate its performance, the time latency of the MVN was evaluated using an internal program that processed data via Ethernet and measured the interval for reading protocol messages at 40 ms to assess data throughput. After receiving protocol messages from different clients, MVN pushed them to a queue where it was expected that all messages would be stored within 40 ms. The computational efficiency of MVN was tested by timing the processing speed of the onboard algorithm until it sent actuation control commands to CAN. To ensure proper

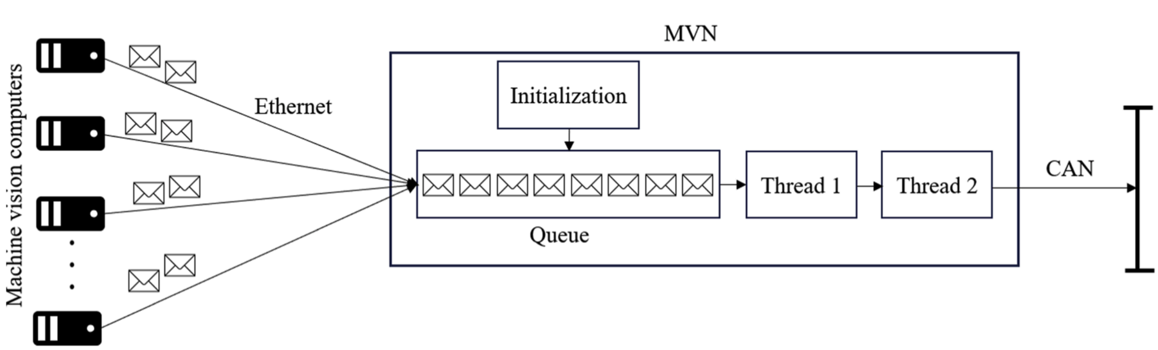


Fig. 4. MVN queue generation for protocol messages received for machine vision computer, following FIFO.

CAN frames were generated, an oscilloscope (DS1102E, Rigol Technology Limited, Beijing, China) was used to observe the signal waveforms from the MVN to assess the signal density and communication timing. The impact of the MVN on the total busload of the ISOBUS network was analyzed using a USB CAN adapter (PCAN-USB, PEAK-System Technik GmbH, Darmstadt, Germany) and its associated software (PCAN-View, V.5.3.0.942, PEAK-System Technik GmbH, Darmstadt, Germany) focusing on busload variations during normal and peak data transmission phases.

Three ISOBUS networks were set up to test the MVN representing different sprayer manufacturers but with unified communication. After the industry-wide adaptation of ISOBUS, manufacturers developed VT for general acceptance across manufacturers and AEF named it universal terminal (UT). The first setup (Fig. 5a) involved an ISOBUS network controlled by a RAVEN UT (CR7, Raven Industries Inc., South Dakota, United States). The second setup (Fig. 5b) involved a custom-made static sprayer with a Raven rate controller and a VT (AFS-Pro 700, Raven Industries Inc., South Dakota, United States). The third setup (Fig. 5c) used a John Deere 4640 UT, specially designed for older John Deere tractors and mixed fleets. Each device operated on a CAN bus system set to a 250 kbps communication speed, adhering to ISOBUS standards to ensure generalization.

2.4. Field apparatus and scenarios

The first field testing involved a full-size boom sprayer (Patriot 3240, Case IH, Wisconsin, USA) with a separate functional spraying system, including two tanks and pumps (Shurflo 2088, Pentair plc, London, UK). This setup was mounted with an AFS Pro-700 VT, the Pentair rate controller, and 60 Pentair rotating valves (Hypro ProStop-E Single, Pentair plc, London, United Kingdom) with nozzles (3D90, Syngenta, Cambridge, United Kingdom) positioned parallel to the original boom sprayer at a 55° angle spraying backward with the travel direction. Fig. 6 provides a visualization of the components of all the setups in terms of their connections on the communication buses. The MVN was tested using 8 machine vision computers (Nuvo-7160GC, Neousys Technology Inc., New Taipei City, Taiwan), each configured to send protocol messages with four different camera IDs, simulating the operation of four connected cameras detecting targets and communicating as actual machine vision systems. In total, the 8 machine vision computers transmitted the equivalent of 30 camera messages simultaneously over the Ethernet network, replicating a full-scale machine vision setup for testing purposes. The MVN generated the CAN frames and transmitted them to the parallel implement bus, i.e., the Pentair system via the CAN 1 channel. Simultaneously, the MVN received speed data from the original implement bus of the sprayer via the CAN 0 channel, enabling it to adjust the nozzle opening duration accordingly.

In the second field test scenario, a complete machine vision system to detect weed called hair fescue was comprised of six OAK-D cameras (OAK-D, OpenCV, Cyprus) mounted 0.5 m above the ground. This setup controlled individual nozzles mounted on a utility task vehicle (UTV) equipped with a Pentair spraying system featuring 12 nozzles (Fig. 7). The angled nozzles sprayed backward similar to the Patriot 3240 setup to enable timely targeting of the weeds. For field experiments, multiple colour printouts of 3 photos of hair fescue tufts (Fig. 8) were placed in clear sheet protectors (28.4 cm x 23.5 cm Standard Clear Sheet Protector, Staples, Canada) as target weeds in real-time.

Using this configuration, experiments were conducted to explore various target distribution patterns simulating common weed distributions in agricultural fields. These patterns included individual targets, intersections (objects located between two cameras FOV), sequential activations (nozzles were activated one after another), wide/bulk distributions (clusters of targets), and elongated targets (aligned with the direction of the UTV), as shown in Fig. 9.

The experiment took place in an open testing area at the Dalhousie Agricultural Campus (Truro, Nova Scotia, Canada) and involved three different speeds: Speed 1 (3.22 kph), Speed 2 (6.44 kph), and Speed 3 (9.66 kph). These speeds represented typical operational ranges for UTV across blueberry terrain allowing us to evaluate the MVN functionalities in establishing successful communication between the machine vision system and the ISOBUS to hit the target accurately.

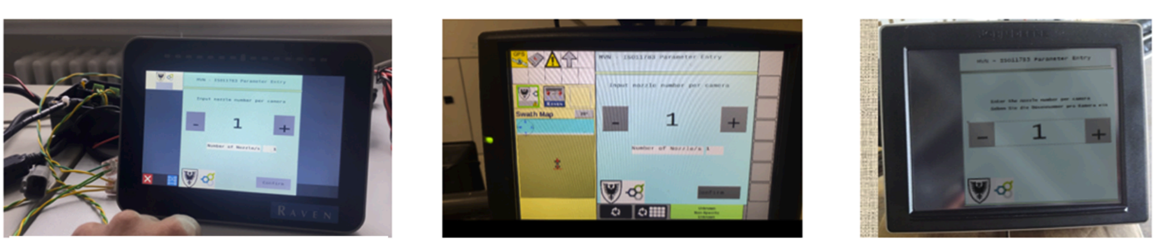
3. Result and discussions

3.1. MVN algorithm performance

The MVN time latency was evaluated across three phases focusing on its ability to handle high frequency data and ensure timely responses in precision spraying tasks. During the Ethernet communication phase, 30 protocol messages were processed within 40 ms demonstrating the capacity of the system to handle multiple simultaneous data streams from 8 computers without significant latency. Here queue management played a crucial role in maintaining real-time data flow.

In the second phase, the MVN successfully read and processed all protocol messages queued within the 40 ms window demonstrating that the memory resources were sufficient to handle bursts of incoming data as depicted in Fig. 10. In the third phase, the overall processing time was evaluated from receiving the protocol messages to generating and transmitting CAN frames. The MVN took 14.3 ms to process the protocol data into a 60-nozzle array and an additional 1.7 ms to create and send four CAN frames to the network. The total time of 16 ms for the complete operation ensured that the system remained responsive and stayed well within the targeted 40 ms window. This time was further optimized using the message queue, threaded programs, and memory management.

The oscilloscope waveform in Fig. 11 illustrates the CAN signal voltage level generated by the MVN. The signal alternates between the dominant voltage level of 3.5 V and the recessive voltage level of 2.5 V. The interval between successive pulses is measured at approximately 4.4 μs corresponding to a frequency of 69.4 kHz, highlighting the high frequency and dense signal transmission. This consistent timing demonstrates the ability of MVN to efficiently generate and transmit CAN frames, enabling faster data communication. The observed waveform reflects the capability of the MVN to handle higher data throughput with stable and precise signal generation, which is key for real-time applications.



(a) (b) (c)

Fig. 5. Screen view while MVN connected with three VT across different manufacturers: (a) RAVEN CR7 UT, (b) Raven AFS-Pro 700 VT, and (c) John Deere 4640 UT.

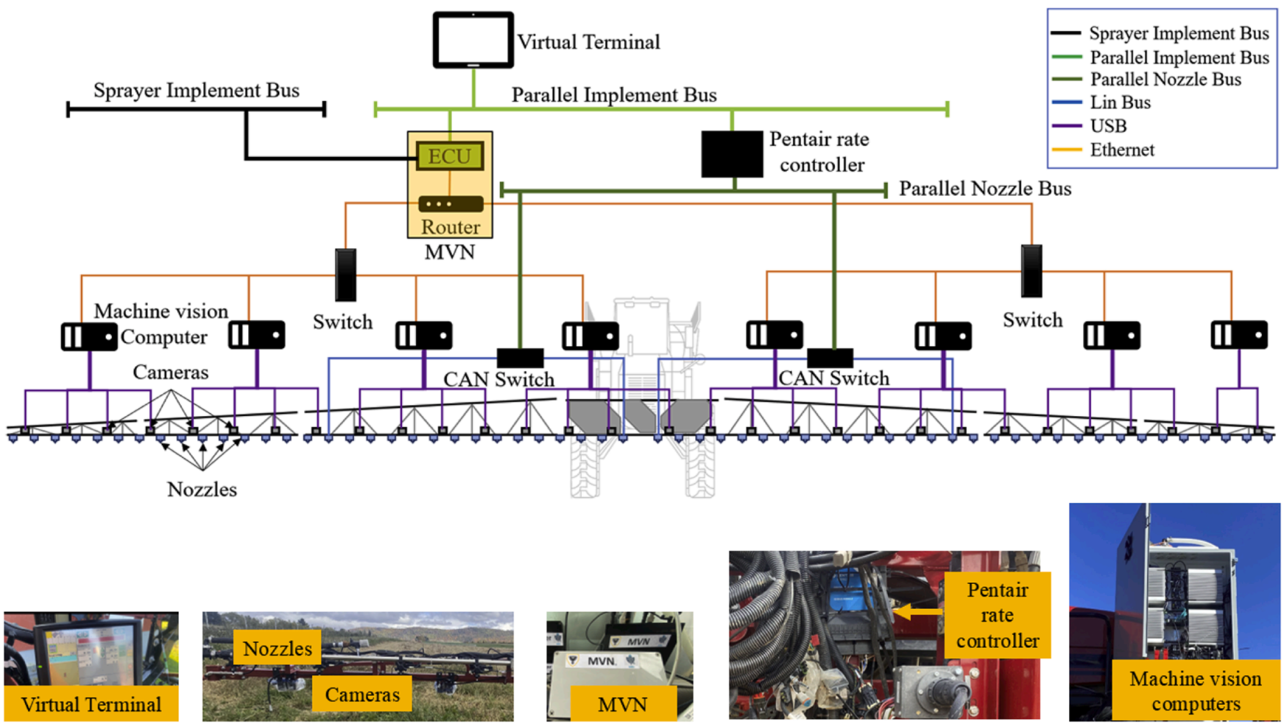


Fig. 6. Drawing of the CAN-compatible spraying system for the Patriot 3240, MVN mounted and integrated into a Pentair spraying system, controlling 60 nozzles individually.

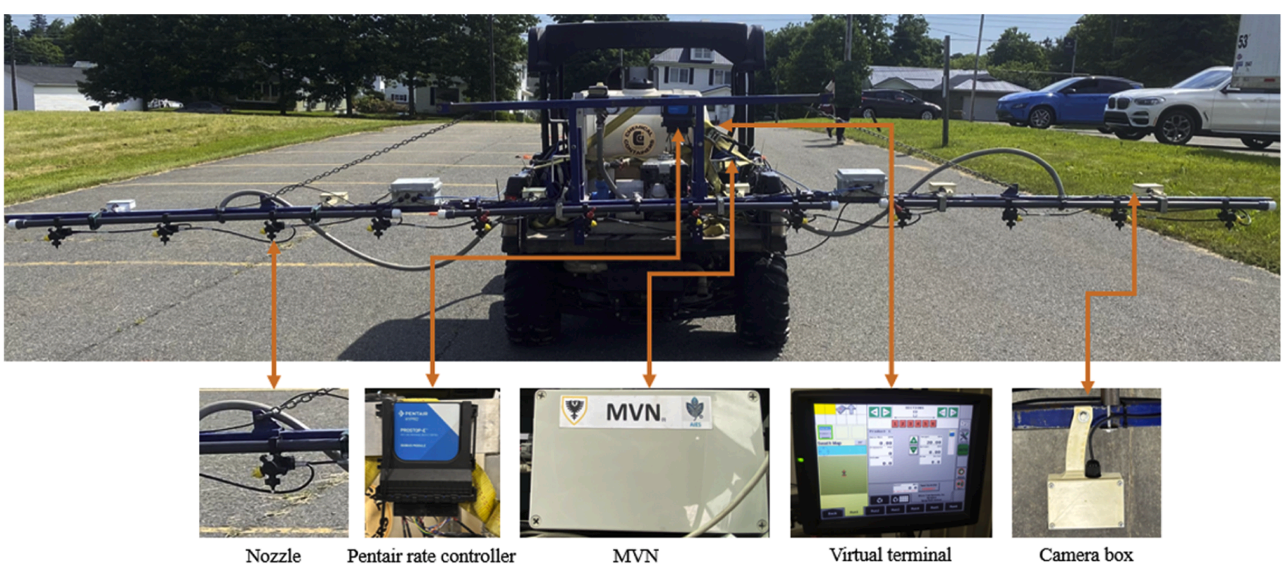


Fig. 7. MVN mounted on UTV, equipped with Pentair spraying system, controlling 12 nozzles through a machine vision system comprising 6 cameras.

When the MVN was tested across three different ISOBUS VT/UT setups, it was consistently connected and displayed successfully. Table 1 summarizes the time it took for each VT/UT to display the interactive screen. These results confirmed that the MVN could connect and display successfully across different VT/UT systems, with varying times for the display to pop up depending on the VT/UT, confirming its compatibility with ISOBUS.

3.2. Busload analysis

The CAN busload performance evaluation and comparison of the algorithms depending on the frequency and increased CAN frames to control a higher number of nozzles are shown in Fig. 12. In this figure, ISOBUS handshaking messages were excluded to ensure a consistent basis for comparison focusing solely on actuation CAN frames over 10 min. When sending one CAN ID every 400 ms the average busload was 0.15 %, which increased to 1.47 % when the CAN ID was sent every 40 ms.

The highest bus load was observed when the algorithm generated four CAN IDs within 40 ms to manage the control of 60 nozzles. This configuration resulted in a significantly higher busload of 5.86 % indicating the increased traffic on the CAN bus due to the higher quantity of CAN frames required to control a greater number of nozzles. This signified the possible minimum busload added to the network to control

Fig. 8. Three different images of Hair Fescue for field experimental detection with UTV setup.

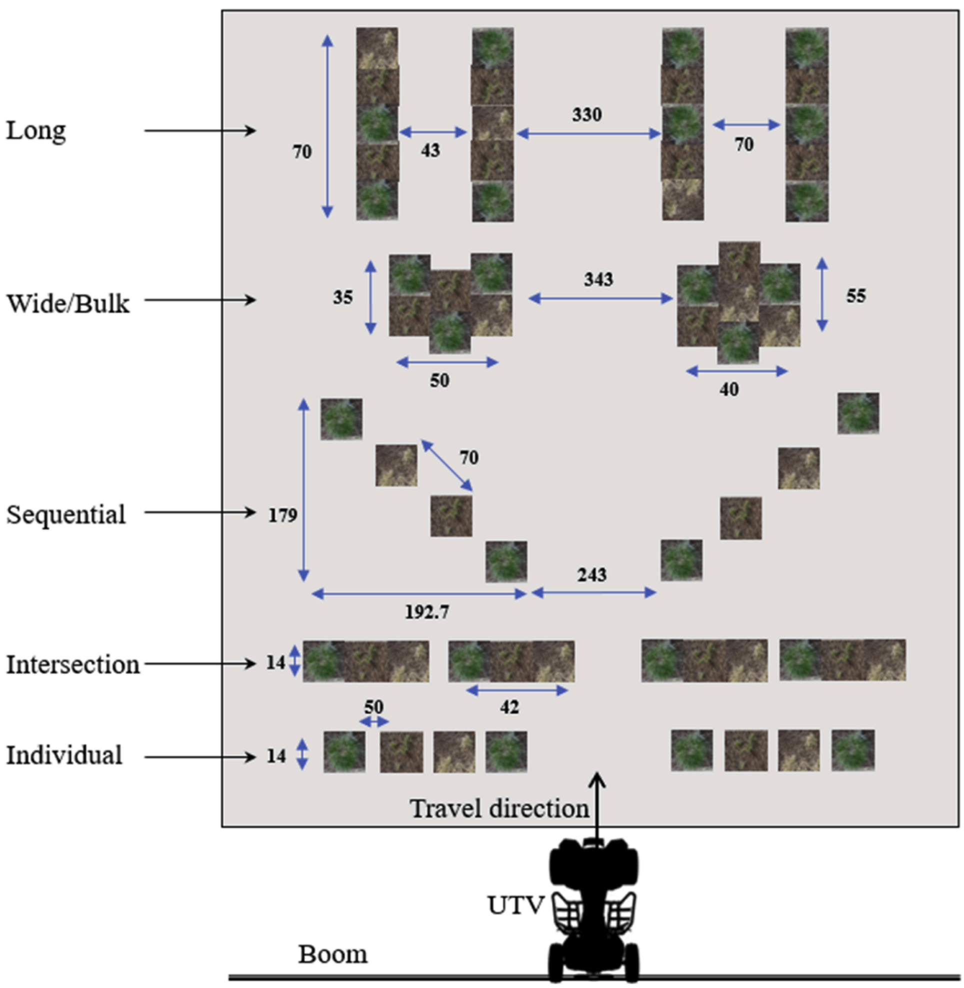


Fig. 9. Target distribution on the ground, showing distances in cm, for the MVN field test.

up to 64 nozzles, highlighting the trade-off between message frequency and busload.

The graphs in Fig. 13 represent the busload performance for 10-min evaluation of the two different spraying systems tested in this research work. The 60-nozzle system on Patriot 3240, showed a bus load that peaked at the initialization at approximately 25 % and before dropping to approximately 22 % at steady state. While the baseline busload was measured at 16.1 % at steady state due to all components broadcasting on the network, integrating the MVN added 5.86 %. The MVN successfully controlled all nozzles processing 450,000 protocol messages that were randomly generated from the 8 machine vision computers and transmitted over Ethernet. It handled a minimum of 30 protocol messages within every 40 ms interval achieving a 100 % success rate in real-time nozzle control.

On the other hand, the busload of the 12-nozzle system on the UTV, where the Pentair rate controller and VT initially occupied 7.44 % of the busload. Adding the MVN increased the busload to 13.30 %. Also in this experiment, the MVN successfully controlled the sprayer, responding to

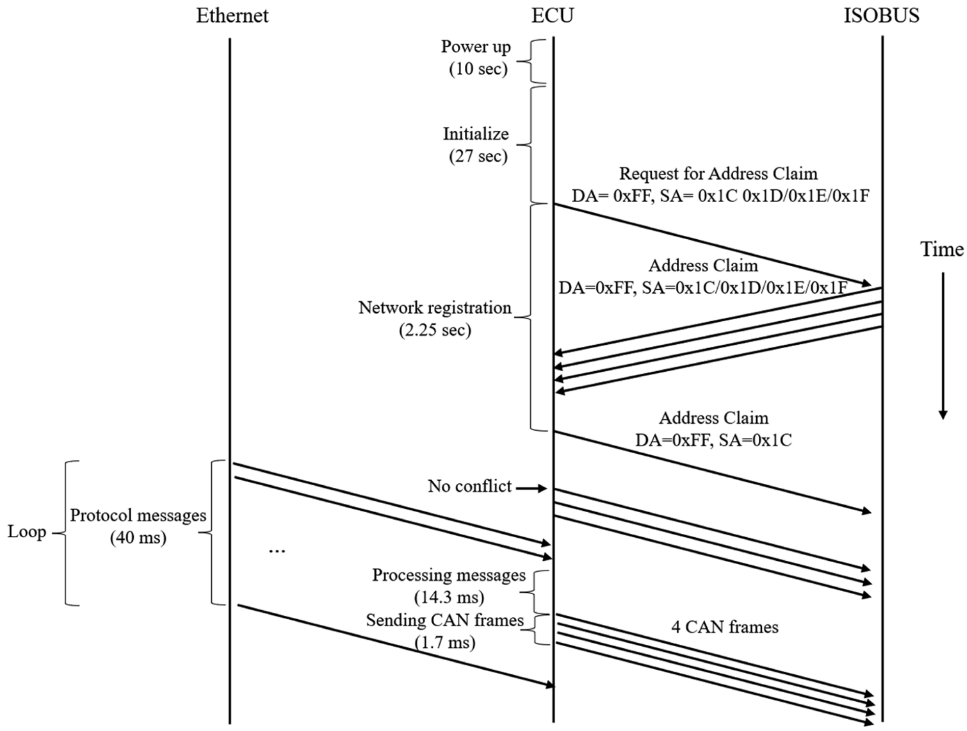


Fig. 10. MVN timeline performance showing Ethernet communication, message processing, and CAN frame transmission.

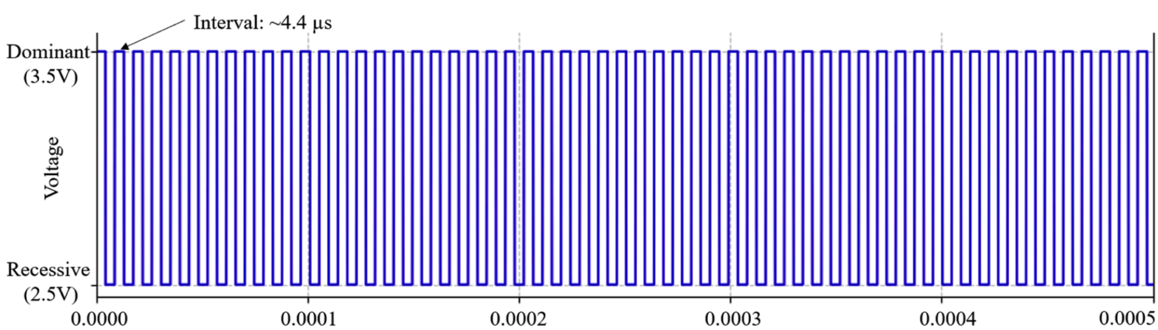


Fig. 11. Oscilloscope signals of CAN frame transmission the MVN.

Table 1
Time taken for MVN to display on various VT/UT across different setups.

VT/UT	With implement?	Displayed on VT?	Time to display (s)
Raven CR7	No	Yes	58
Raven AFS PRO700	Yes	Yes	51.5
John Deere 4640	No	Yes	70

6 protocol messages within 40 ms with a 100 % success rate. For both cases, the implement bus load remained within the acceptable 45 % limit.

3.3. Field implementation analysis

Table 2 summarizes the results of running the UTV three times at each speed while spraying on targets with different distributions. In the final field experiment, the MVN adjusted the nozzle opening time based on UTV speed, ensuring consistent spray length across all speeds. The goal was to maintain uniform spray coverage regardless of vehicle speed by varying the nozzle opening time. At Speed 1, for both individual and intersection targets, a buffer spray of approximately 28.76 cm was applied on each side. This buffer remained consistent at higher speeds, with nozzle opening time reduced for Speed 2 and Speed 3. Despite this reduction, the spray length remained the same for individual and intersection targets across all speeds, demonstrating the MVN's success in adjusting for speed changes.

For larger target distributions, such as sequential, bulk, and long, the MVN dynamically adjusted the nozzle opening time to maintain consistent spray lengths across all speeds. This confirmed the algorithm's ability to control nozzle timing and maintain uniform coverage.

While the dynamic nozzle adjustment improved spot spraying precision, it led to a fixed spray length, potentially causing slightly higher spray coverage than desired. Additionally, spray density varied inversely with speed—slower speeds resulted in denser spray, while faster speeds resulted in less dense spray. This variation presents a potential drawback to achieving consistent spray density across different speeds, even when the nozzle opening time is adjusted. Future improvements could focus on addressing this issue by controlling the pressure and flow rate on the boom, ensuring uniform spray density regardless of speed. However, compared to a uniform distribution approach this method significantly reduced overall spray usage.

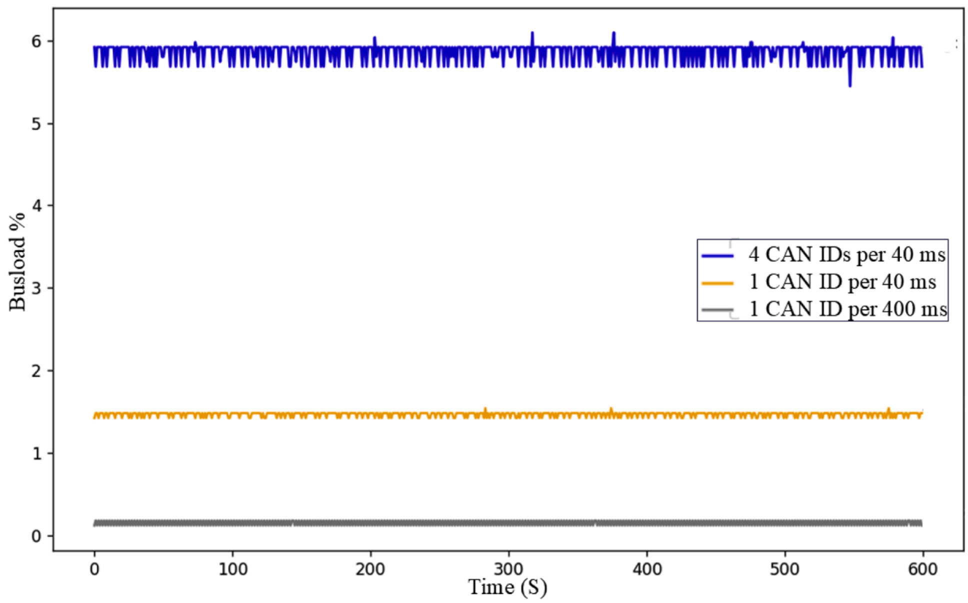


Fig. 12. MVN busload performance comparison at different numbers of CAN frames and their frequency.

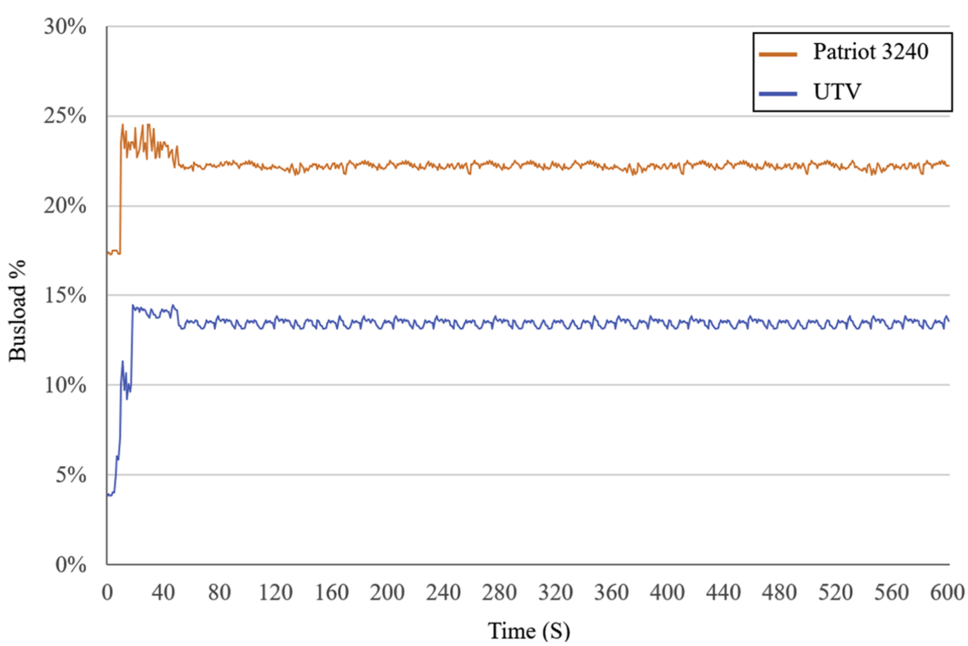


Fig. 13. ISOBUS load performance during a 10-min evaluation of the two different spray systems, on Patriot 3240, and the UTV.

Table 2
Unified spraying areas at different distributions of objects and different speeds.

Distribution Type	Target Area Length (cm)	Speed 1	Speed 2	Speed 3
		Spray length (cm)		
Individual	14	71.52	71.52	71.52
Intersection	14	71.52	71.52	71.52
Sequential	179	329	329	329
Bulk	35	95	95	95
Long	70	200	200	200

4. Conclusion

The novel ISOBUS-compliant MVN introduced in this study was able to integrate multiple machine vision systems to wide boom sprayers, enabling individual nozzle control for real-time spot spraying on different targets through hybrid communication. The MVN was capable of communicating with both the implement bus and machine vision systems, efficiently managing high-frequency protocol messages with minimal latency. This was achieved by incorporating a client-server threaded program, along with checksum verification and message queuing. The ability to dynamically adjust nozzle opening times based on sprayer speed ensured consistent spray coverage, reducing overspray and optimizing resource use. The MVN maintained a high but uniform busload distribution, staying within the standard busload threshold. The key findings highlight the capability of the MVN to handle a minimum of 30 messages within 40 ms via the Ethernet protocol used in the system processes to control 60 individual nozzles. The ability of MVN to adapt to real-time speed variations optimized spray usage and ensured coverage of all detected targets.

The compliance of MVN with ISOBUS systems across various VTs demonstrated its versatility and adaptability in different setups, highlighting its potential for widespread adoption in precision agriculture. Future improvements could focus on further field trials in actual spraying using pesticides for the final evaluation of performance on wide boom sprayers. The MVN offers a scalable solution for precision agriculture, capable of converting real-time spot spraying of pesticides on wide boom sprayers based on machine vision into reality.

Ethics statement

Not applicable: This manuscript does not include human or animal research.

CRediT authorship contribution statement

Mozammel Bin Motalab: Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation, Conceptualization. **Ahmad Al-Mallahi:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Alex Martynenko:** Writing – review & editing, Supervision. **Karama Al-Tamimi:** Writing – review & editing, Supervision. **Dimitrios S. Paraforos:** Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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